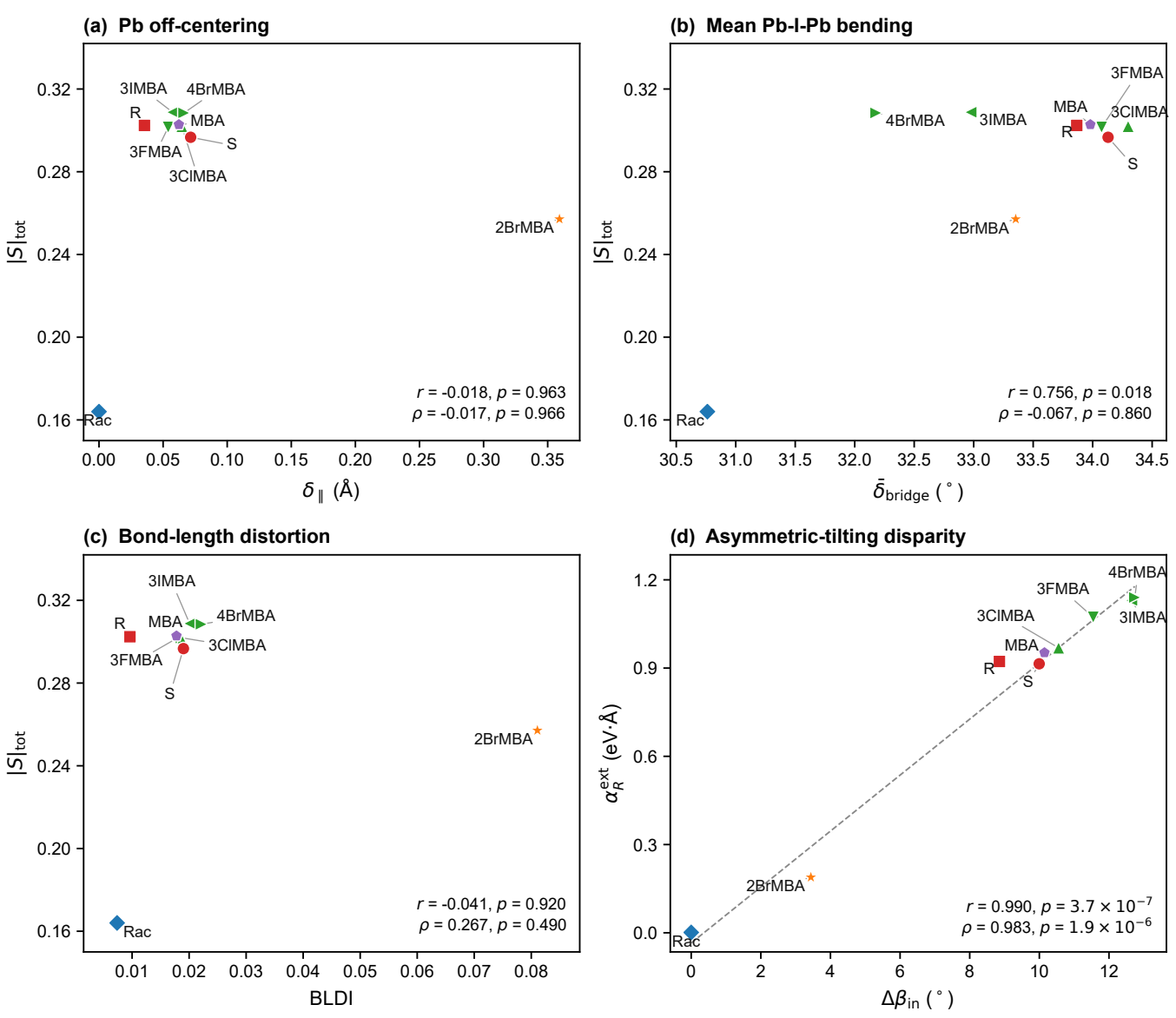

(a) Pb off-centering
3IMBA
4BrMBA
R
MBA
S
3FMBA
3ClMBA
2BrMBA
Rac
r = -0.018, p = 0.963
ρ = -0.017, p = 0.966
|S|tot
δ∥ (Å)
(b) Mean Pb-I-Pb bending
3FMBA
4BrMBA
3IMBA
MBA
3ClMBA
R
S
2BrMBA
Rac
r = 0.756, p = 0.018
ρ = -0.067, p = 0.860
|S|tot
δ̄bridge (°)
(c) Bond-length distortion
3IMBA
R
MBA
4BrMBA
3ClMBA
3FMBA
S
2BrMBA
Rac
r = -0.041, p = 0.920
ρ = 0.267, p = 0.490
|S|tot
BLDI
(d) Asymmetric-tilting disparity
4BrMBA
3FMBA
3ClMBA
3IMBA
MBA
R
S
2BrMBA
Rac
r = 0.990, p = 3.7 × 10−7
ρ = 0.983, p = 1.9 × 10−6
α_R^ext (eV·Å)
Δβin (°)